



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Course Code: CSA1703

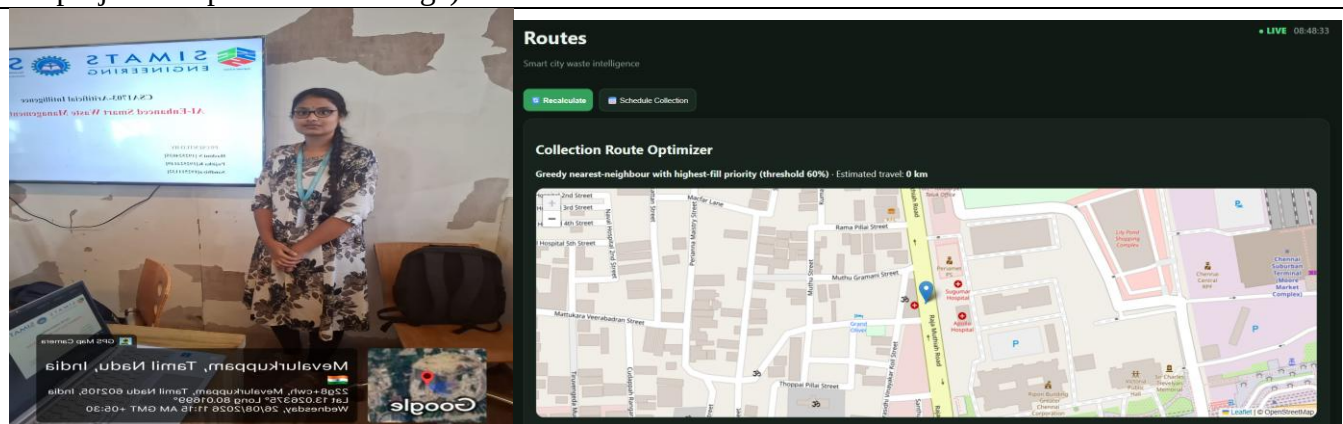
Slot: A

Course Name: ARTIFICIAL INTELLIGENCE

Course Faculty: Dr. R JEENA

Project Title: AI-Enhanced Smart Waste Management System

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

The AI-Enhanced Smart Waste Management System is an intelligent solution designed to improve the efficiency of waste collection and monitoring. Traditional waste collection mainly depends on fixed schedules, which may result in unnecessary collection trips or overflowing bins. The proposed system uses IoT sensors, Artificial Intelligence, data analysis, and real-time monitoring to overcome these problems.

The system continuously monitors the fill level of waste bins using sensors. The collected data is transmitted to a central platform for processing and analysis. AI techniques are then used to analyse historical and real-time waste data to predict future waste accumulation. Based on the predicted fill level, the system can identify bins that are likely to become full and generate collection alerts.

The system classifies waste bins into different conditions such as Normal, Warning, and Critical. When a bin reaches a predefined threshold, such as 80% capacity, a warning is generated.

Student Signature
P ROHINI

Guide Signature